

Lab - DLM for prescribed kinematics in Navier-Stokes equations

Ivan Fumagalli and Nicola Parolini

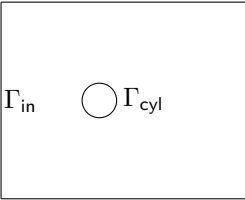
MOX, Dipartimento di Matematica, Politecnico di Milano

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PhD Course - *Numerical Methods for Multi-Physics Problems in Moving Domains*

Navier-Stokes problem

Find the solution $\mathbf{u} : \Omega \times (0, 5] \rightarrow \mathbb{R}^d$ and $p : \Omega \times (0, 5] \rightarrow \mathbb{R}$ such that

$$\left\{ \begin{array}{ll} \rho \partial_t \mathbf{u} + \rho \mathbf{u} \cdot \nabla \mathbf{u} - \nu \Delta \mathbf{u} + \nabla p = \mathbf{f} & \text{in } \Omega, t \in (0, 5] \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, t \in (0, 5] \\ \mathbf{u} = \mathbf{u}_{\text{in}} & \text{on } \Gamma_{\text{in}}, t \in (0, 5] \\ \boldsymbol{\sigma} \mathbf{n} = \mathbf{0} & \text{on } \Gamma_{\text{out}}, t \in (0, 5] \\ \mathbf{u} = \mathbf{0} & \text{on } \Gamma_w \cup \Gamma_{\text{cyl}}, t \in (0, 5] \\ \mathbf{u} = \mathbf{0} & \text{in } \Omega, t = 0 \end{array} \right.$$


with the settings of

https://wwwold.mathematik.tu-dortmund.de/~featflow/en/benchmarks/cfdbenchmarking/flow/dfg_benchmark1_re20.html.

Starting from the template files

TEMPLATE_vanDANaprescribedKinematics.py,
common/TEMPLATE_fluid_variational_problem.py,
common/TEMPLATE_lagrange_variational_problem.py,
solve the problem using the DLM method.

DLM method - splitting scheme

Given $\mathbf{u}_h^0 \in V_h$ and $\lambda_h^0 \in \Lambda_h$, interpolate $\lambda_h^0 \in \Lambda_h$ to $\hat{\lambda}_h^0 \in V_h$, then for each $n = 0, 1, \dots$:

- ① find $\mathbf{u}_h^* \in V_h$, $p_h^{n+1} \in Q_h$ s.t., for all $\mathbf{v}_h \in V_h$, $q_h \in Q_h$,

$$\begin{aligned} a^{n+1}(\mathbf{u}_h^*, \mathbf{v}_h) + b(p_h^{n+1}, \operatorname{div} \mathbf{v}_h)_\Omega - b(q_h, \operatorname{div} \mathbf{u}_h^*)_\Omega \\ = L^{n+1}(\mathbf{v}_h) - (\hat{\lambda}_h^n, \mathbf{v}_h)_\Omega \end{aligned} \quad (\text{DLM-NS-S1})$$

- ② interpolate $\mathbf{u}_h^* \in V_h$ to $\tilde{\mathbf{u}}_h^* \in \Lambda_h$

- ③ find $\lambda_h^{n+1} \in \Lambda_h$ s.t., for all $\zeta_h \in \Lambda_h$,

$$(\lambda_h^{n+1}, \zeta_h)_{\Omega_t^s} = \frac{1}{\Delta t} (\zeta_h, \tilde{\mathbf{u}}_h^* - \bar{\mathbf{u}}^{n+1})_{\Omega_t^s} + (\lambda_h^n, \zeta_h)_{\Omega_t^s} \quad (\text{DLM-NS-S2})$$

- ④ interpolate $\lambda_h^{n+1} \in \Lambda_h$ to $\hat{\lambda}_h^{n+1} \in V_h$

- ⑤ find $\mathbf{u}_h^{n+1} \in V_h$ s.t., for all $\mathbf{v}_h \in V_h$,

$$\left(\frac{\rho}{\Delta t} \mathbf{u}_h^{n+1}, \mathbf{v}_h \right)_\Omega = \left(\frac{\rho}{\Delta t} \mathbf{u}_h^*, \mathbf{v}_h \right)_\Omega + (\hat{\lambda}_h^{n+1} - \hat{\lambda}_h^n, \mathbf{v}_h)_\Omega \quad (\text{DLM-NS-S3})$$

One can also iterate these steps, to improve the enforcement of the constraint.

DLM method - splitting scheme - vanDana

We can push the splitting even further, considering a projection method (based on the PISO algorithm) to solve (DLM-NS-S1) at each time t^{n+1} in three sub-steps [vanDANA manual].

Given $\mathbf{u}_h^n \in V_h$, $p_h^n \in Q_h$ and $\hat{\boldsymbol{\lambda}}_h^n \in V_h$:

- a find $\mathbf{u}_h^{**} \in V_h$ s.t., for all $\mathbf{v}_h \in V_h$,

$$a^{n+1}(\mathbf{u}_h^{**}, \mathbf{v}_h) = L^{n+1}(\mathbf{v}_h) + \langle \hat{\boldsymbol{\lambda}}_h^n, \mathbf{v}_h \rangle - b(p_h^n, \operatorname{div} \mathbf{v}_h) \quad (\text{PISO-1})$$

- b find $p_h^* \in Q_h$ s.t., for all $q_h \in Q_h$,

$$(\nabla p_h^*, \nabla q_h)_\Omega = (\nabla p_h^n, \nabla q_h)_\Omega - \frac{\rho}{\Delta t} b(q_h, \operatorname{div} \mathbf{u}_h^{**}) \quad (\text{PISO-2})$$

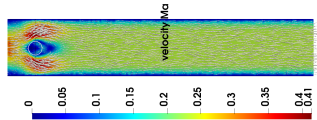
- c find $\mathbf{u}_h^* \in V_h$ s.t., for all $\mathbf{v}_h \in V_h$,

$$\frac{\rho}{\Delta t} (\mathbf{u}_h^*, \mathbf{v}_h)_\Omega = \frac{\rho}{\Delta t} (\mathbf{u}_h^{**}, \mathbf{v}_h)_\Omega - (\nabla(p_h^* - p_h^n), \mathbf{v}_h)_\Omega \quad (\text{PISO-3})$$

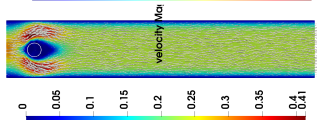
One can also iterate steps b-c, to improve the accuracy in the solution of (DLM-NS-S1).

Numerical results - $t = 0.6$

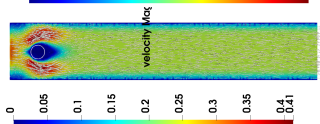
DLM iter = 1
 $\Delta t = 0.1$



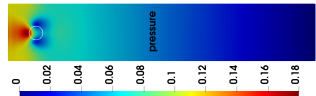
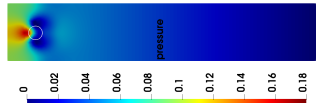
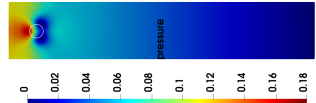
DLM iter = 1
 $\Delta t = 0.01$



DLM iter = 10
 $\Delta t = 0.1$

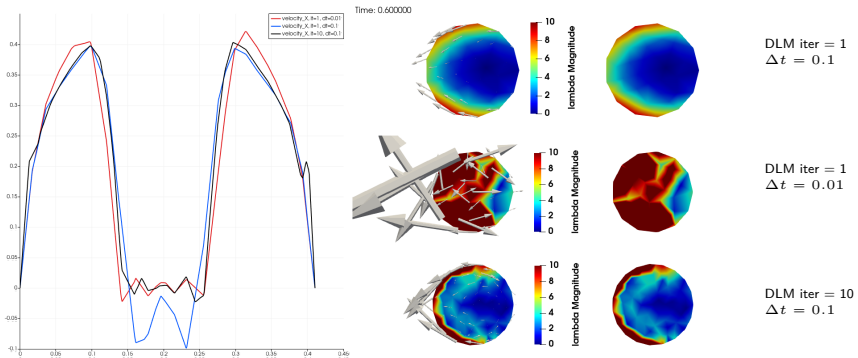


Time: 0.00000



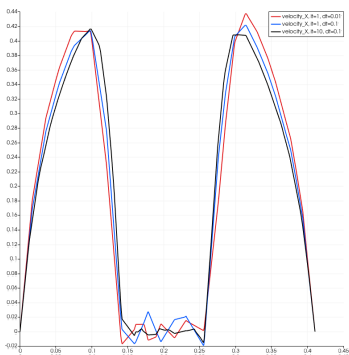
Velocity and pressure plots show slight differences close to the obstacle.

Numerical results - $t = 0.6$

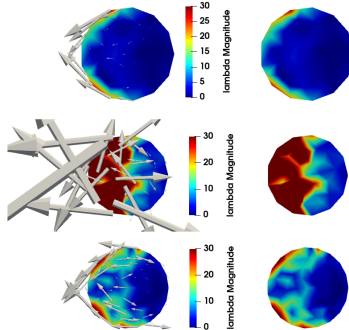


Velocity profile shows that fulfillment of constraint $\mathbf{u} = \mathbf{0}$ in Ω^s improves by reducing Δt or sub-iterating the DLM splitting steps. The Lagrange multiplier plot can be interpreted as the force acting on the fluid particles to stop them and make them satisfy the constraint.

Numerical results - $t = 5.0$



Time: 5.000000



DLM iter = 1
 $\Delta t = 0.1$

DLM iter = 1
 $\Delta t = 0.01$

DLM iter = 10
 $\Delta t = 0.1$

For long times, the constraint is satisfied in all cases. Yet, the Lagrange multiplier still presents some differences.

The Paraview state to analyze the results is on WeBeep:
`cylComparison.pvsm`